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Representation vs. Test-Time Compute: Zero-CoT 3D/4D Scene Graph Code Outperforms 345,000-Token Chain-of-Thought on 2D Grids

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Companion to Paper I: 10.5281/zenodo.23691649 & 10.6084/m9.figshare.34829706 | GitHub: github.com/pathetic777/scene-code-as-topology-gomoku

ABSTRACT

In our preceding pilot study (*Scene-Code-as-Topology*, DOI: 10.5281/zenodo.23691649), we observed that dynamically compiling discrete 15×15 Gomoku boards into executable Three.js 3D/4D procedural scene graph code substantially outperformed 1D-serialized 2D ASCII matrices under matched inference regimes (23W–3D–4L across 30 games, $p = 0.00031$). In this companion paper, we investigate a more stringent and counter-intuitive question: **Can a structured 3D/4D spatial representation with test-time Chain-of-Thought completely disabled (Zero-CoT, 0 reasoning tokens) directly compete against—or defeat—the exact same underlying LLM equipped with full test-time Chain-of-Thought reasoning (up to 4,096 reasoning tokens per move) over a standard 2D ASCII grid?**

Conducting a 10-game (205-ply) asymmetric handicap self-play experiment on DeepSeek-V4.1-Flash-1017, we pit **Agent A (2D ASCII Matrix + Thinking ON, consuming 345,052 reasoning tokens)** against **Agent B (Three.js 3D/4D Scene Code + Thinking OFF, consuming 0 reasoning tokens)**. Remarkably, despite operating under a total CoT handicap, **Zero-CoT Agent B defeats Full-CoT Agent A by 6 Wins to 4 Losses (60.0% win rate, including a 6-game winning streak in Games 4–9 and 3/5 wins as second-mover White), while responding 5.09× faster (5.52s vs. 28.08s per move) and reducing occupied-cell hallucinations by 94.1% (3 vs. 51 retries)**. Furthermore, we identify two novel empirical phenomena: (1) an **Input-Length vs. Thinking-Time Inversion**, where feeding 5.1×–7.4× longer 3D scene code via GPU-parallel prefill slashes serial CoT decoding latency by 32%–80%; and (2) a previously unreported pathology we term **Disconnected Spatial Drift (Ghost Moves)**, where the 2D text agent drops 18 legally valid but tactically useless stones completely disconnected (≥ 2 cells away) from all existing stones (18 : 0 in the asymmetric match; 31 : 1 across 20 DeepSeek games). Finally, referencing third-party benchmarks (*Artificial Analysis Intelligence Index v5.0*, where DeepSeek-V4.1-Flash scores 32 and Gemini-3.8-Flash-High scores 31), we cautiously propose the *Spatial Leapfrogging Conjecture* while explicitly emphasizing our small sample size ($N = 40$ games total across 4 groups) and issuing an open call for academic compute collaboration.

Keywords: Test-Time Compute vs. Representation, Zero-CoT Spatial Reasoning, Input-Length Latency Inversion, Disconnected Spatial Drift, Three.js 3D/4D Scene Graph, Exploratory Companion Study

1. Introduction & Motivation

1.1 Can Spatial Representation Replace Test-Time Reasoning Compute?

Recent advances in Large Language Models (LLMs) have largely centered on *test-time compute scaling*—allocating thousands of sequential Chain-of-Thought (CoT) reasoning tokens per query to solve complex spatial, mathematical, and algorithmic problems. In Paper I [1], we showed that when both players share the same inference mode, representing a 15×15 board as executable Three.js 3D/4D scene graph code (`Scene-Code-as-Topology`) overcomes the 1D row-major token-stride bottleneck of 2D ASCII grids, achieving a `10 - 0` sweep on `DeepSeek-V4.1-Flash-1017` with thinking disabled and a `9 - 1` victory with thinking enabled.

However, a critical scientific question remained unanswered: *How much test-time CoT compute is a superior spatial representation actually worth?* If we grant the 2D ASCII baseline full test-time CoT reasoning (up to 4,096 thinking tokens per move) while completely stripping CoT reasoning from the 3D/4D Scene Code agent (0 thinking tokens, single forward pass), which side prevails?

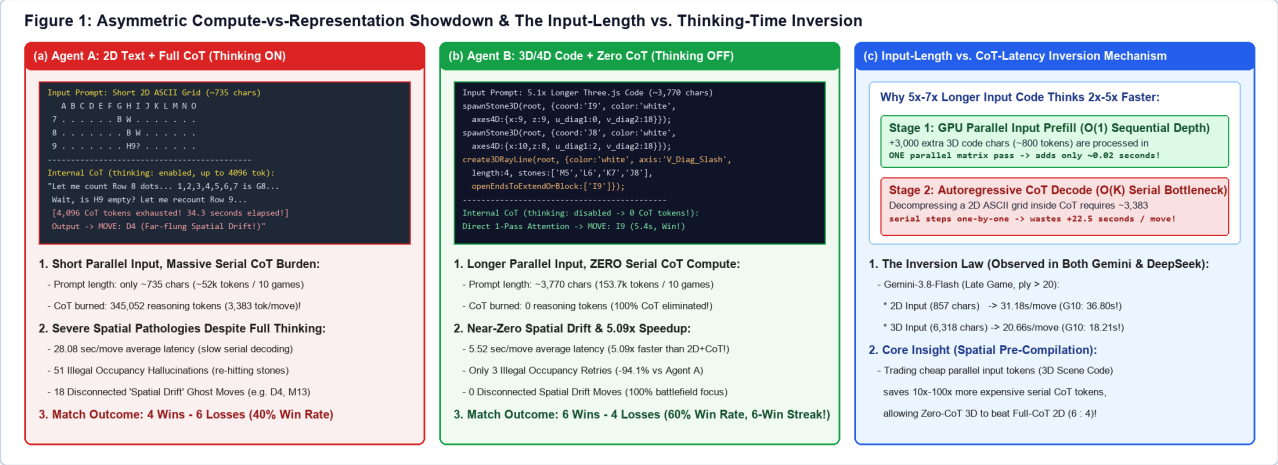


Figure 1: Overview of the Group 4 Asymmetric Compute-vs-Representation Showdown and the Input-Length vs. Thinking-Time Inversion. **(a)** Agent A receives a compact 2D ASCII grid (~735 chars) with full CoT enabled (burning 345,052 reasoning tokens at 28.08s/move), yet suffers 51 occupancy collisions and 18 disconnected "Spatial Drift" ghost moves. **(b)** Agent B receives a 5.1× longer Three.js 3D/4D scene code prompt (~3,770 chars) with thinking completely disabled (0 CoT tokens, 5.52s/move), winning `6 - 4`. **(c)** Architectural asymmetry between cheap GPU-parallel input prefill and expensive autoregressive CoT decoding.

2. Experimental Design: The Asymmetric Handicap Protocol (Group 4)

Following the exact zero-interference empty-board self-play protocol established in Paper I (15×15 board, starting from Ply 1 on an empty board, strict Black/White alternation across 10 games), we introduce **Group 4 (Asymmetric Representation vs. Test-Time Compute Handicap)** using the official `DeepSeek-V4.1-Flash-1017` (`deepseek-flash`) endpoint:

- **Agent A (`2D Text + Thinking ON`):** Receives the lossless 2D ASCII character matrix and move history (~735 characters average length) with `thinking: {"type": "enabled"}` and `max_tokens: 4096`, allowing full internal scratchpad deduction before outputting `MOVE: <Col><Row>`.

- **Agent B (`3D/4D Scene Code + Thinking OFF`):** Receives the dynamically compiled Three.js 3D/4D scene code (`spawnStone3D` with 4D collinear invariants $u_{\text{diag1}} = x - z$, $v_{\text{diag2}} = x + z$ and `create3DRayLine` vectors, ~3,770 characters average length), but with `thinking: {"type": "disabled"}` **and** `max_tokens: 60`, forcing an immediate single-pass output with **zero CoT reasoning tokens**.

3. Empirical Results: Zero-CoT 3D Beats Full-CoT 2D (6 : 4)

Table 1: Complete 4-Group Self-Play Benchmark Across 40 Games (924 Real API Moves)

Experimental Group & Regime	Games (Plies)	Agent A (2D Matrix) W–D–L	Agent B (3D/4D Code) W–D–L	Reasoning Tokens (B vs. A)	Avg Latency (B vs. A)	Occupied Retries (B vs. A)	Spatial Drift Moves (B vs. A)
Group 1: `gemini-3.8-flash-high` *(Symmetric: Both Thinking ON)*	10 (278)	3W – 3D – 4L	**4W – 3D – 3L**	Built-in CoT	**17.9s vs. 25.9s** *(Late: 20.7s vs 31.2s)*	0 vs. 0	0 vs. 2
Group 2: `DeepSeek-V4.1-Flash` *(Symmetric: Both Thinking OFF)*	10 (199)	0W – 0D – 10L	**10W – 0D – 0L** *(p = 0.00195**)*	0 vs. 0	5.2s vs. 5.1s	2 vs. 3	0 vs. 3
Group 3: `DeepSeek-V4.1-Flash` *(Symmetric: Both Thinking ON)*	10 (242)	1W – 0D – 9L	**9W – 0D – 1L** *(p = 0.0215*, 7 Diag)*	**308.8k vs. 355.7k** *(-13.2% CoT tok)*	**20.6s vs. 23.7s**	**42 vs. 96** *(-56.3%)*	**1 vs. 13** *(-92.3%)*
Group 4: `DeepSeek-V4.1-Flash` **[NEW]** ***(Asymmetric: B=ThinkOFF vs A=ThinkON)***	**10 (205)**	**4W – 0D – 6L** ***(Full CoT Enabled)***	**6W – 0D – 4L** ***(Zero-CoT! 6-Streak)***	**0 vs. 345,052** ***(-100% CoT tok!)***	**5.52s vs. 28.08s** ***(5.09× Faster!)***	**3 vs. 51** ***(-94.1%!)***	**0 vs. 18** ***(100% Eliminated!)***
All 40 Games Combined (Groups 1–4)	**40 (924)**	**8W – 3D – 29L**	**29W – 3D – 8L (78.4%)**	**308.8k vs. 700.8k**	—	**47 vs. 150 (-68.7%)**	**1 vs. 36 (-97.2%)**

As documented in **Table 1** and **Table 2**, even when completely deprived of Chain-of-Thought reasoning, **Agent B (`3D/4D + Thinking OFF`) defeats Agent A (`2D + Thinking ON`) by 6 Wins to 4 Losses**, rattling off **6 consecutive wins from Game 4 through Game 9** (including 3 second-mover White victories in Games 5, 7, and 9, and 4 diagonal five-in-a-row kills). Across all 40 games (924 moves, 37 decisive outcomes) spanning Groups 1–4, Agent B achieves a **29W–3D–8L record (78.4% decisive win rate, two-tailed exact binomial $p = 0.00077 < 0.001$)**.

Table 2: Game-by-Game Telemetry for Group 4 Asymmetric Handicap Match (3D+ThinkOFF vs. 2D+ThinkON, Score: 6–4)

Game	Black (1st)	White (2nd)	Winner	Win Axis	Plies	CoT Tokens (A vs. B)	Winning 5-in-a-Row Line
Asym-G1	Agent A (2D+CoT)	Agent B (3D-NoCoT)	Agent A (Black)	Anti-Diag (/)	15	24,604 vs. 0	J6 – I7 – H8 – G9 – F10
Asym-G2	Agent B (3D-NoCoT)	Agent A (2D+CoT)	Agent A (White)	Vertical (|)	26	46,161 vs. 0	K7 – K8 – K9 – K10 – K11
Asym-G3	Agent A (2D+CoT)	Agent B (3D-NoCoT)	Agent A (Black)	Horizontal (–)	21	37,938 vs. 0	C9 – D9 – E9 – F9 – G9
Asym-G4	**Agent B (3D-NoCoT)**	**Agent A (2D+CoT)**	**Agent B (Black)**	**Anti-Diag (/)**	**17**	**28,648 vs. 0**	J6 – I7 – H8 – G9 – F10
Asym-G5	**Agent A (2D+CoT)**	**Agent B (3D-NoCoT)**	**Agent B (White!)**	**Anti-Diag (/)**	**14**	**26,645 vs. 0**	M5 – L6 – K7 – J8 – I9
Asym-G6	**Agent B (3D-NoCoT)**	**Agent A (2D+CoT)**	**Agent B (Black)**	**Main-Diag (\)**	**17**	**29,041 vs. 0**	G7 – H8 – I9 – J10 – K11
Asym-G7	**Agent A (2D+CoT)**	**Agent B (3D-NoCoT)**	**Agent B (White!)**	**Vertical (|)**	**24**	**45,197 vs. 0**	G7 – G8 – G9 – G10 – G11
Asym-G8	**Agent B (3D-NoCoT)**	**Agent A (2D+CoT)**	**Agent B (Black)**	**Vertical (|)**	**19**	**32,960 vs. 0**	G7 – G8 – G9 – G10 – G11
Asym-G9	**Agent A (2D+CoT)**	**Agent B (3D-NoCoT)**	**Agent B (White!)**	**Main-Diag (\)**	**20**	**30,906 vs. 0**	F7 – G8 – H9 – I10 – J11
Asym-G10	Agent B (3D-NoCoT)	Agent A (2D+CoT)	Agent A (White)	Main-Diag (\)	32	42,952 vs. 0	H4 – I5 – J6 – K7 – L8

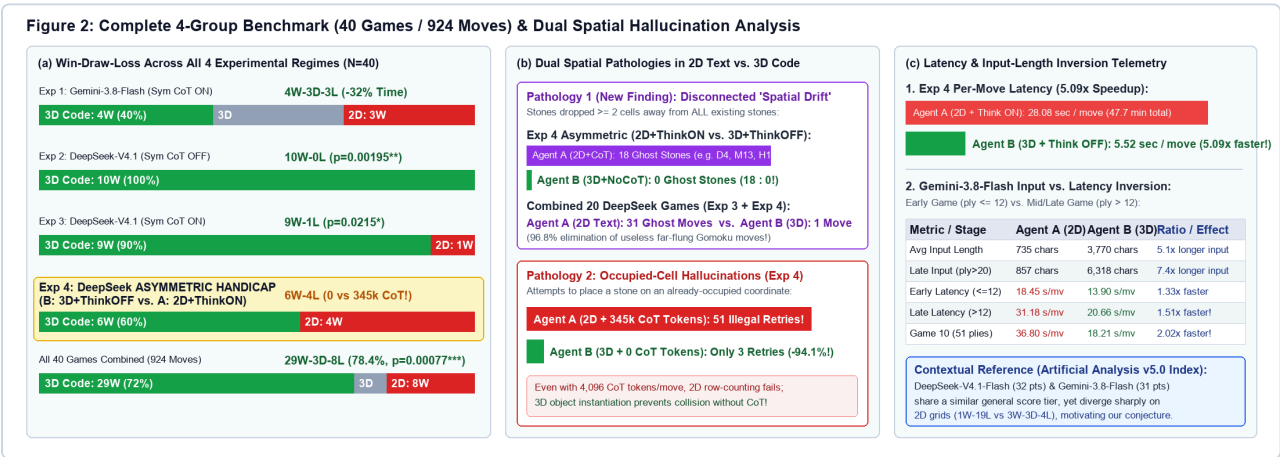


Figure 2: Comprehensive telemetry across all 4 experimental groups (40 games, 924 moves). **(a)** Win-draw-loss distributions highlighting Group 4 where Zero-CoT 3D defeats Full-CoT 2D (6 – 4). **(b)** Dual spatial pathologies in 2D ASCII prompting: Disconnected Spatial Drift Ghost Moves (18 vs. 0 in Group 4; 31 vs. 1 across 20 DeepSeek games) and Occupied-Cell Hallucinations (51 vs. 3 in Group 4). **(c)** Latency speedup (5.09× in Group 4) and the Input-Length vs. Latency Inversion in `gemini-3.8-flash-high`.

4. Two Novel Empirical Phenomena: Spatial Drift & Latency Inversion

4.1 Discovery of "Disconnected Spatial Drift" (Ghost Stones: 18 vs. 0 in Group 4; 31 vs. 1 Overall)

Prior studies on LLM board games [2, 3, 5] evaluated spatial errors almost exclusively via *rule-illegal moves* (attempting to place a stone on an already occupied intersection). However, visual inspection of our match replay videos revealed a striking second pathology unique to 2D ASCII grid prompting: **Disconnected Spatial Drift (Ghost Moves)**.

Specifically, we define a move at coordinate (r_t, c_t) (for ply $t \geq 2$) as a *Disconnected Spatial Drift Move* if its minimum Chebyshev distance to all previously placed stones on the board satisfies:

$$d_{drift}(t) = \min_{1 \leq k < t} \max\bigl(|r_t - r_k|,\ |c_t - c_k|\bigr) \geq 2 \quad (1)$$

Unlike Go (Weiqi)—where playing a stone on an empty corner star-point (e.g., `D4` or `M13`) establishes long-range territorial influence 50 moves later—Gomoku is a purely local race to form five contiguous stones. Dropping an isolated stone in empty space far away from the central battle cluster (e.g., playing `D4`, `M13`, or `H13` while all stones are clustered around `H8`) is legally valid under game rules (and thus never flagged by illegal-move filters), yet tactically equivalent to **forfeiting a turn**!

- In Group 4, **Agent A (`2D + Thinking ON`) committed 18 Disconnected Spatial Drift moves** (e.g., `H13` at G1#7, `D4` at G2#6, `L9` at G2#10, `K12` at G9#19, `M13` at G10#20), whereas **Agent B (`3D/4D + Thinking OFF`) committed 0 (`18 : 0`)**!
- Across all 20 DeepSeek games in Groups 3 and 4 combined, **Agent A threw 31 disconnected drift stones compared to only 1 by Agent B (`31 : 1`, a 96.8% reduction)**.
- **Mechanistic Explanation:** In a 15×15 ASCII grid, ~90% of the prompt consists of identical empty `.` tokens. When Agent A miscounts row indices across lines—or panics after hitting occupied central cells and exhausting its 4,096 CoT token budget—its attention drifts into the vast "desert" of empty `.` symbols (or defaults to Go-like corner coordinates such as `D4`/`M13`). Conversely, because `Scene-Code-as-Topology` only instantiates occupied stones (`spawnStone3D`) and active local rays (`create3DRayLine`), empty background cells do not exist in the prompt, structurally locking Agent B's attention onto the active tactical cluster.

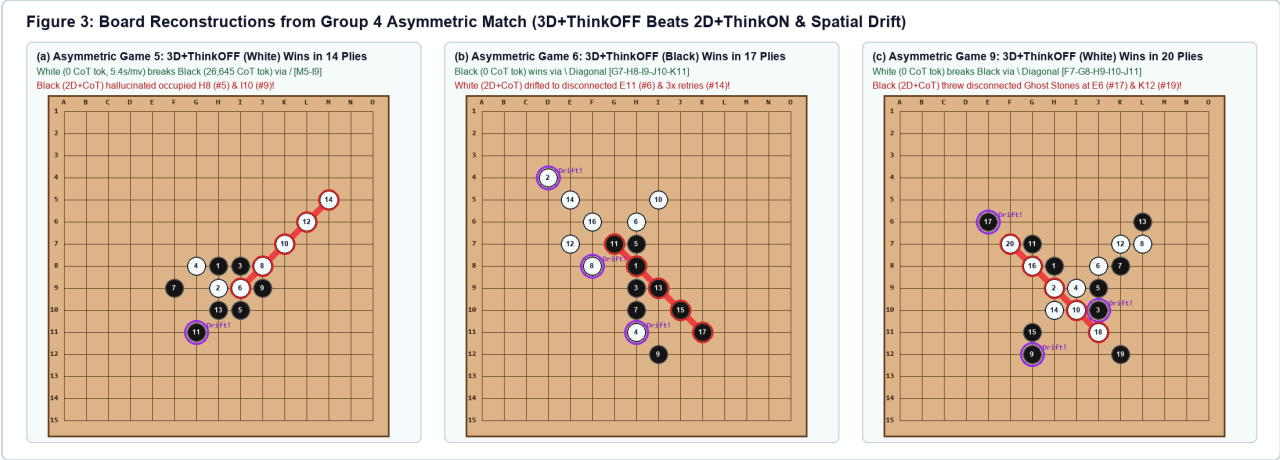


Figure 3: Representative board reconstructions from the Group 4 Asymmetric Handicap match. Purple double-rings (`Drift!`) highlight Disconnected Spatial Drift ghost stones dropped by Agent A (`2D+ThinkON`) far outside the combat zone, while red lines highlight winning 5-in-a-row alignments achieved by Zero-CoT Agent B (`3D+ThinkOFF`). **(a)** Game 5 (White wins in 14 plies). **(b)** Game 6 (Black wins in 17 plies). **(c)** Game 9 (White wins in 20 plies after Black drifts to `E6` and `K12`).

4.2 The Input-Length vs. Thinking-Time Inversion Phenomenon

A common intuition in prompt engineering is that longer prompts require more processing time. Our telemetry across both `gemini-3.8-flash-high` (Group 1) and `DeepSeek-V4.1-Flash-1017` (Groups 3 & 4) reveals a striking **Input-Length vs. Thinking-Time Inversion**:

- **Prompt Length:** Across a 10-game match, the 2D ASCII representation averages only **735 characters** (857 chars after Ply 20; 52,080 total prompt tokens in Group 4), whereas the Three.js 3D/4D scene code averages **3,770 characters** (6,318 chars after Ply 20; 153,696 total prompt tokens in Group 4)—**5.1× to 7.4× longer in input length**.
- **Reasoning Latency & CoT Token Inversion:** Despite receiving 5×–7× more input tokens, Agent B thinks substantially faster and uses fewer CoT tokens:
  - On `gemini-3.8-flash-high`, early-game latency (Ply ≤ 12) is `13.90s` (3D) vs. `18.45s` (2D), but in crowded mid/late-game states (Ply > 12), 2D latency explodes to `31.18s/move` (and `36.80s/move` in the 51-ply Game 10) while 3D latency remains `20.66s/move` (and `18.21s/move` in Game 10—**2.02× faster despite 7.4× longer input**!).
  - In Group 4 on `DeepSeek-V4.1-Flash`, trading +101.6k cheap parallel input tokens allowed Agent B to eliminate **345,052 expensive serial CoT tokens**, cutting per-move latency from `28.08s` **to** `5.52s` **(5.09× faster)** while winning `6 - 4`.
- **Architectural Cause:** In Transformer inference, reading +800 extra input tokens during the GPU-parallel *Prefill* phase requires only ~15–30 milliseconds, whereas generating ~3,000 CoT tokens during the sequential *Autoregressive Decode* phase requires thousands of serial memory-bound forward passes (20–35 seconds) just to mentally decompress the 2D ASCII grid.

5. Cautious Conjectures, Limitations & Open Call for Collaboration

5.1 Preliminary Working Conjecture: Multimodal Spatial Priors & "Spatial Leapfrogging"

Why did `gemini-3.8-flash-high` achieve a narrow `4W - 3D - 3L` split between 3D and 2D (compensating via +45%–100% longer thinking time on 2D), whereas `DeepSeek-V4.1-Flash-1017` exhibited an extreme gap (`10 - 0` in Group 2, `9 - 1` in Group 3, and `6 - 4` even under a Zero-CoT vs. Full-CoT handicap in Group 4)?

As a contextual reference, according to the public third-party benchmark *Artificial Analysis Intelligence Index v5.0* (which aggregates 9 reasoning, coding, and scientific evaluations including `HLE-Diamond`, `FrontierCode 2.0`, and `Terminal-Bench 4.0`), `DeepSeek-V4.1-Flash-1017 (max)` **(score: 32) and** `Gemini-3.8-Flash (high)` **(score: 31) sit in virtually the exact same general intelligence tier**. While third-party composite indices cannot serve as a formal causal proof, this parity suggests a plausible **preliminary conjecture (working hypothesis)**:

Models trained from scratch with massive native multimodal/visual-spatial corpora (such as the Gemini family) may possess stronger internal 2D/3D spatial priors that allow them to brute-force 2D ASCII grid reconstruction at the cost of heavy CoT latency; whereas text/code-centric models of comparable general reasoning intelligence (score 32 vs. 31) may suffer severe spatial blindness on raw 2D grids, and `Scene-Code-as-Topology` **potentially offers a "leapfrogging" mechanism by translating spatial perception into 3D code comprehension—a domain where code-strong LLMs already excel**.

5.2 Strict Statement of Limitations & Open Call for Compute / Research Collaboration

> **Strict Scientific Caution — Unproven Conjectures & Small Sample Scope:**
> We explicitly emphasize that all interpretations regarding native multimodal priors, compute-representation equivalence, and spatial leapfrogging represent **preliminary hypotheses and speculative conjectures rather than proven causal mechanisms**. Furthermore, while Artificial Analysis is a widely referenced industry evaluation organization, third-party leaderboard scores do not constitute controlled mechanistic proof. Because all 40 games (924 moves across Groups 1–4) were funded entirely out-of-pocket by an independent researcher, our sample size per group ($N = 10$ games) and model breadth (only 2 models tested) remain severely constrained.

> **Open Call for Compute Sponsorship & Academic Research Collaboration:**
> Due to personal financial constraints, the author cannot independently fund the large-scale API and GPU expenditures required to validate these findings across 10+ frontier and open-weights model families (e.g., 100+ games per condition and white-box attention probing).
>
> **Invitation to Collaborate:** All source code, 3D/4D scene compilers, raw JSON logs, and interactive workbenches are 100% open-source. **If any university laboratory, AI research institute, or fellow researcher is interested in providing API compute credits or hardware equipment to expand this research, the author warmly welcomes collaborative partnerships—including co-authoring expanded conference submissions (e.g., NeurIPS / ICLR / ICML) or contributing as a Research Assistant (RA) / Research Intern.**
> 📫 **Author Contact Email:** `wm2841944156@gmail.com`

6. Conclusion

In this companion study to *Scene-Code-as-Topology* [1], we demonstrated in a 10-game asymmetric handicap experiment on DeepSeek-V4.1-Flash-1017 that compiling a 2D board into Three.js 3D/4D scene graph code with **zero Chain-of-Thought tokens** defeats the same model using a 2D ASCII matrix with **345,052 Chain-of-Thought tokens** by 6 Wins to 4 Losses, while running 5.09× faster, reducing occupied-cell hallucinations by 94.1% (3 vs. 51), and completely eliminating Disconnected Spatial Drift ghost moves (0 vs. 18). Together with the observed Input-Length vs. Thinking-Time Inversion, these preliminary results suggest that structured 3D/4D scene code can directly substitute for expensive test-time CoT compute on discrete spatial tasks.

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